

ArmBench in the robot-policy adaptation landscape

Runtime reliability and statistical evidence are distinct contributions from supervised fine-tuning, RL adaptation, and human-in-the-loop learning.

METHOD / STATUS	ADAPTATION LOCUS	LEARNING SIGNAL	LATENCY QUESTION	EVIDENCE DOMAIN
ArmBench PROJECT EVIDENCE NOT PEER REVIEWED	Runtime dispatcher Frozen official pi0.5	No training	Skip delay-matched prefix; execute the following suffix	LIBERO simulation 4 suites / 40 tasks paired + manifest-bound
RTC NEURIPS 2025 FORMAL PAPER	Inference-time action-chunk inpainting	No retraining	Freeze committed actions; inpaint the continuation	Dynamic simulation + real bimanual tasks
OpenVLA-OFT RSS 2025 FORMAL PAPER	VLA weight fine-tuning	Offline imitation / L1 regression	Faster parallel decoding; async staleness not central	LIBERO simulation + real bimanual ALOHA
DPPO ICLR 2025 FORMAL PAPER	Diffusion-policy RL fine-tuning	Policy gradients / reward	Not an asynchronous runtime method	Simulation + zero-shot hardware deployment
HIL-SERL SCIENCE ROBOTICS 2025 FORMAL PAPER	Real-world policy learning	Demos + human corrections + RL	Not an action-chunk latency method	Real robot + dexterous tasks

Solid border: formal venue/journal paper Dashed border: project evidence, not a publication

ARMBENCH TODAY

Strong systems evidence, limited policy-learning claim

- Frozen official pi0.5 checkpoint
- No fine-tuning or hidden task tuning
- Frozen protocols + matched conditions
- Exact tests, multiplicity, bootstrap CI
- Source/checkpoint/video attestation

TOP-CONFERENCE GAP

- Measured jitter and wall-clock deadlines
- Same-checkpoint head-to-head with RTC
- More than one VLA checkpoint
- Real-robot closed-loop validation
- Mechanism beyond fixed suffix skipping

These gaps test generality and deployment realism;
they do not imply that RL is mandatory.

NEAREST SCIENTIFIC COMPARATOR

RTC: training-free, asynchronous action-chunk
execution with simulation and real-robot evidence.